(See extended delta §3 for details)

Recall: $f[-MX^{m_M}]\cdot 1 = \sigma^m f(x)$ (See Prop 3.5.1 below)

where σ is a linear operator on $\Lambda_{\mathbb{K}}$ s.t. $\sigma \tilde{H}_\mu = t^{n(\mu)} q^{n(\mu^*)} \tilde{H}_\mu$, $n(\gamma) = \sum_{i=1}^{\ell(\gamma)} (i-1)\gamma_i$.

We denote the Weyl symmetrization operator for GL_ℓ by

$$\sigma(f(z_1,\dots,z_\ell)) = \sum_{w\in S_\ell} w\left(\frac{f(z)}{\prod_{\alpha\in R_+}(1-z^{-\alpha})}\right).$$

(See extended delta §3.4 for examples)

- Fix a weight ρ s.t. $\langle \alpha_i^\vee, \rho\rangle = 1$ $\forall$ simple coroot $\alpha_i^\vee$ (e.g. $\rho=(\ell-1,\ell-2,\dots,1,0)$) ($\rho$: unique up to adding a constant vector)
  - If $\mu\in\mathbb{Z}^\ell$ s.t. $\mu+\rho$ is not regular (i.e. with equal parts), then $\sigma(x^\mu)=0$.
  - If $\mu\in\mathbb{Z}^\ell$ s.t. $\mu+\rho$ is regular, then for $w\in S_\ell$ s.t. $w(\mu+\rho)-\rho$ is dominant, then $\sigma(x^\mu)=(-1)^{\ell(w)}\chi_{w(\mu+\rho)-\rho}$

    That is why we have $+(2,1,0)$

    e.g. • $\mu=(1,2,0)$ and $\mu+(2,1,0)=(3,3,0)$ $\quad \sigma(x^\mu) = \sum_{w\in S_3} w\left(\frac{x_1^3x_2^3\cdot x_1^1x_2^2x_3^0}{(x_1-x_2)(x_1-x_3)(x_2-x_3)}\right)=0$

    • $\mu=(1,3,0)$ and $\mu+(2,1,0)=(3,4,0)$

    Then $w=s_1$. Hence $\sigma(x^{1,3,0}) = \sum_{w\in S_3} w\left(\frac{x_1^2x_2\cdot x_1^3x_2^4x_3^0}{(x_1-x_2)(x_1-x_3)(x_2-x_3)}\right) = \sum_{w\in S_3} w\left(\frac{x_1^3x_2^4x_3^0}{(x_1-x_2)(x_1-x_3)(x_2-x_3)}\right) = \sum_{w\in S_3}(-1)\, w\left(\frac{x_1^4x_2^3x_3^0}{(x_1-x_2)(x_1-x_3)(x_2-x_3)}\right) = (-1)\sigma(x^{2,2,0})$

Let $\eta(z_1,\dots,z_\ell)\in\mathbb{K}[z_1^{\pm1},\dots,z_\ell^{\pm1}]$ be a Laurent polynomial. A q,t raising operator series is an expression of the form

$$h(z) = \sigma\left(\frac{\eta(z)}{\prod_{\alpha\in R_+}(1-qz^\alpha)\prod_{\alpha\in R_+}(1-tz^\alpha)}\right)$$

interpreted as an infinite formal linear combination of GL_ℓ characters χ_λ with coefficients in $\mathbb{K}$ by expanding $\frac{1}{(1-qz^\alpha)} = 1+qz^\alpha+q^2z^{2\alpha}+\cdots$ and $\frac{1}{1-tz^\alpha} = 1+tz^\alpha+t^2z^{2\alpha}+\cdots$ before applying σ.

Note that there are only finitely many $\mu\in\mathbb{Z}^\ell$ s.t. $\sigma(x^\mu)=\pm\chi_\lambda$ and hence computing it in this way makes sense

(expand $\frac{1}{1-qz^\alpha}$, $\frac{1}{1-tz^\alpha}$ in geometric series first before applying σ.)

e.g. $\ell=3$, $\eta(z)=\frac{1}{qt}\frac{z_2}{z_3}+\frac{t}{q^2}\cdot\frac{1}{z_1^2z_2}$.

Then

$$h(z)=h(z_1,z_2,z_3)=\sigma\left(\frac{\frac{1}{qt}\frac{z_2}{z_3}+\frac{t}{q^2}\frac{1}{z_1^2z_2}}{\left(1-q\frac{z_1}{z_2}\right)\left(1-q\frac{z_1}{z_3}\right)\left(1-q\frac{z_2}{z_3}\right)\left(1-t\frac{z_1}{z_2}\right)\left(1-t\frac{z_1}{z_3}\right)\left(1-t\frac{z_2}{z_3}\right)}\right)$$

$$=\sigma\left(\left(\frac{1}{qt}\frac{z_2}{z_3}+\frac{t}{q^2}\cdot\frac{1}{z_1^2z_2}\right)\left(1+q\frac{z_1}{z_2}+q^2\frac{z_1^2}{z_2^2}+\cdots\right)\left(1+q\frac{z_1}{z_3}+q^2\frac{z_1^2}{z_3^2}+\cdots\right)\left(1+t\frac{z_1}{z_2}+t^2\frac{z_1^2}{z_2^2}+\cdots\right)\left(1+t\frac{z_1}{z_3}+t^2\frac{z_1^2}{z_3^2}+\cdots\right)\left(1+t\frac{z_2}{z_3}+t^2\frac{z_2^2}{z_3^2}+\cdots\right)\right)$$

$$=\sigma\left(\frac{1}{qt}\frac{z_2}{z_3}+\frac{t}{q^2}\frac{1}{z_1^2z_2^2}+\frac{1}{t}\frac{z_1}{z_3}+\frac{t}{q}\cdot\frac{1}{z_1z_2^2}+\cdots\cdots\right)$$

$$=\frac{1}{qt}\cancel{\sigma(z^{0,1,1})}^{\,0}+\frac{t}{q^2}\cancel{\sigma(z^{2,1,0})}^{\,0}+\frac{1}{t}\underbrace{\sigma(z^{1,0,-1})}_{\chi_{1,0,-1}}+\frac{t}{q}\underbrace{\sigma(z^{1,-2,0})}_{\chi_{-1,-1,-1}}+\cdots$$

If we write $\eta(z)=c(q,t)\,\eta'(z)$ where $\eta'(z)\in\mathbb{Z}[q,t][z_1^{\pm1},\dots,z_\ell^{\pm1}]$, then $h(z)$ can be expressed in a scalar in $\mathbb{K}$ times a power series in q and t over the ring of virtual GL_ℓ characters $\mathbb{Z}[z_1^{\pm1},\dots,z_\ell^{\pm1}]^{S_\ell}$ and this power series expansion is the same as the raising operator series expansion.

e.g. $\ell=3$, $\eta(z)=\frac{1}{qt}\frac{z_2}{z_3}+\frac{t}{q^2}\frac{1}{z_1^2z_2}$. Then $\eta(z)=\overbrace{\frac{1}{q^2t}}^{c(q,t)}\overbrace{\left(q\frac{z_1}{z_3}+t^2\frac{1}{z_1z_2}\right)}^{\eta'(z)}$ which implies $h(z)=\frac{1}{q^2t}\sigma\left(\frac{\eta'(z)}{\underbrace{\prod_{1\le i<j\le3}\left(1-q\frac{z_i}{z_j}\right)\left(1-t\frac{z_i}{z_j}\right)}}\right)$

If we expand this as above as group like-terms, then the coeff. of each virtual character is a $\underbrace{\text{power series in } q \text{ and } t}$

(We did not have that before because we have fractions like $\frac{t}{q}$)

$\therefore$ $h(z)$ considered as a symmetric rational function (i.e. not expanding $\frac{1}{1-qz^\alpha}$, $\frac{1}{1-tz^\alpha}$ and just apply σ) over $\mathbb{K}$ determines its raising operator series expansion.

* Not every symmetric rational function is a raising operator series.

In most of this paper, we regard Catalanimals merely as symmetric rational functions.

Since $H(R_q,R_t,R_{qt},\lambda)=\sigma\left(\frac{z^\lambda\prod_{\alpha\in R_{qt}}(1-qtz^\alpha)}{\prod_{\alpha\in R_q}(1-qz^\alpha)\prod_{\alpha\in R_t}(1-tz^\alpha)}\right)=\sigma\left(\frac{z^\lambda\prod_{\alpha\in R_{qt}}(1-qtz^\alpha)\prod_{\alpha\in R_q\setminus R_{qt}}(1-qz^\alpha)\prod_{\alpha\in R_t\setminus R_{qt}}(1-tz^\alpha)}{\prod_{\alpha\in R_q}(1-qz^\alpha)\prod_{\alpha\in R_t}(1-tz^\alpha)}\right)$, (Take $\eta(z)=z^\lambda\prod_{\alpha\in R_{qt}}(1-qtz^\alpha)\prod_{\alpha\in R_q\setminus R_{qt}}(1-qz^\alpha)\prod_{\alpha\in R_t\setminus R_{qt}}(1-tz^\alpha)\in\mathbb{K}[z_1^{\pm1},\dots,z_\ell^{\pm1}]$)

we can also view Catalanimal as a q,t raising operator series.

The polynomial characters of GL_ℓ are irreducible characters χ_λ for which λ is a partition.

We define $h(z)_{pol}$ of a raising operator series $h(z)$ to be its truncation to polynomial characters. (see path §2 for example)

If $h(z)$ is homogeneous of degree d, then all irreducible characters χ_λ in it have $|\lambda| = d$, so $h(z)_{pol}$ is a finite linear combination of polynomial GL_ℓ characters (schur functions).

i.e. $h(z)_{pol}$ is a symmetric polynomial in $z_1, \dots, z_\ell$ with coefficients in $\mathbb{K}$.

Prop. 3.5.1: Let H be a Catalanimal of length ℓ s.t. $\mathcal{D}_p(H) \in \Lambda(X^{m,1})$ with $\mathrm{Cub}(H) = f$. i.e. $\mathcal{D}_p(H) = f[-MX^{m,1}]$.

Then $(\omega \nabla^m f)(z_1, \dots, z_\ell) = H_{pol}$.

Moreover, this determines $\nabla^m f$ because the Schur expansion $\omega \nabla^m f(x)$ contains only s_λ with $\ell(\lambda) \le \ell$.

Proof: By Lemma 3.4.1 in Path, $f(X^{m,1})$ acts on Λ as $\nabla^{-m} f[-\frac{X}{M}]^\circ \nabla^{-m}$.

$\because \nabla(1) = 1$

$\therefore f(X^{m,1}) \cdot 1 = \nabla^m f[-\frac{X}{M}]^\circ \nabla^{-m} \cdot 1 = \nabla^m f[-\frac{X}{M}]^\circ 1 = \nabla^m \left(f[-\frac{X}{M}] \cdot 1 \right) = \nabla^m \left(f[-\frac{X}{M}] \right)$

$\therefore f[-MX^{m,1}] \cdot 1 = \nabla^m f[-\frac{-MX}{M}] = \nabla^m f(X)$

By Prop 3.5.2 in Path, Choose φ to be the Laurent polynomial $\equiv \Phi(R_{qf}, R_t, R_{qt}, \lambda) \pmod{I^\ell}$, then $\zeta = \mathcal{D}(\Phi) = f[-MX^{m,1}]$ and hence

$$\therefore \omega(\zeta \cdot 1)(z_1, \dots, z_\ell) = H_{qt}(\Phi)_{pol} \Leftrightarrow \omega(f[-MX^{m,1}] \cdot 1)(z_1, \dots, z_\ell) = \sigma\left(\frac{\Phi(R_{qf}, R_t, R_{qt}, \lambda) \prod_{\alpha \in R_{qt}} (1 - q_f t z^\alpha)}{\prod_{\alpha \in R_q} (1 - q_f z^\alpha) \prod_{\alpha \in R_t} (1 - t z^\alpha)} \right)_{pol} = \sigma\left(\frac{z^\lambda \prod_{\alpha \in R_{qt}} (1 - q_f t z^\alpha)}{\prod_{\alpha \in R_q} (1 - q_f z^\alpha) \prod_{\alpha \in R_t} (1 - t z^\alpha)} \right)_{pol}$$

$$\Leftrightarrow \omega(\nabla^m f)(z_1, \dots, z_\ell) = H_{pol}.$$

Moreover, $\omega(\zeta \cdot 1) = \omega(\nabla^m f)$ contains only s_λ with $\ell(\lambda) \le \ell$ and hence determines $\zeta \cdot 1 = \nabla^m f$. $\square$